

Hardware & Budget Report

What to buy, what to rent, and when each one pays for itself

Prepared for Amro · syslab-server project

Compiled 3 September 2026 · Dubai, DEWA commercial electricity rates

Figures drawn from two earlier working sessions: *What The Machine Actually Costs* and *The Bandwidth Wall*, plus the measured serving-throughput benchmarks in the project roadmap. GPU prices moved with a 2026 memory shortage — re-check before committing money.

In brief

- One unified server, multiple cards, serving many clients — not a race to match a frontier model's raw reasoning.
- The honest edge over a cloud API is tool-calling accuracy, latency, cost per token at steady volume, and data that never leaves the building.
- Renting answers "how much do I actually need" for a few hundred dollars before any card is bought.
- Every buy-vs-rent line in this report crosses somewhere between 7 months and 9 years — utilisation decides which side of that line you're on, not the hardware.

What this server is trying to be

The stated aim is one unified server, multiple cards, serving many clients, agentic, "comparable to the power of a cloud AI API," with video optional. That target sets the honest expectation up front: a single on-prem box will not out-reason a frontier model. What it can match or beat is tool-calling accuracy (the best open models now score 92% on the Berkeley Function Calling benchmark), latency (the network hop to a data centre is often the biggest part of a cloud response), throughput per dollar at steady volume, and two things a cloud API structurally cannot offer: data that never leaves the building, and a per-request cost that stops climbing with usage. That is what this budget is being spent on.

Recommended agent model

Model	Berkeley FC	Licence	Note
Qwen3-Coder-30B-A3B-Inst ruct	92%	Apache 2.0	MoE, ~30B total / ~3B active → ~18 GB at Q4, speed of a small model
GLM-4.5-Air	86%	Commercial-frien dly	Second choice if the Qwen family doesn't fit the stack
Llama 3.1 70B	78–84%	Community licence	700M monthly-user cap on the licence
DeepSeek-V3.2	88%	Research-only	Cannot be sold — scores well, ruled out by licence alone

Berkeley Function Calling score is a tool-use accuracy benchmark, not a general reasoning score.

Hardware budget tiers

Budget	Build	What it unlocks
\$0	RTX 3060 12 GB (already owned)	Build and demo only — not a client-facing tier
\$3,000 – 4,000	One RTX 5090 32 GB	The entry point for real clients: 8–12 concurrent
\$7,000 – 10,000	Two RTX 5090s	Agent and image generation on separate cards — lane isolation
\$12,000 – 14,000	One RTX PRO 6000 96 GB	25+ clients, best cost per token, one machine to operate

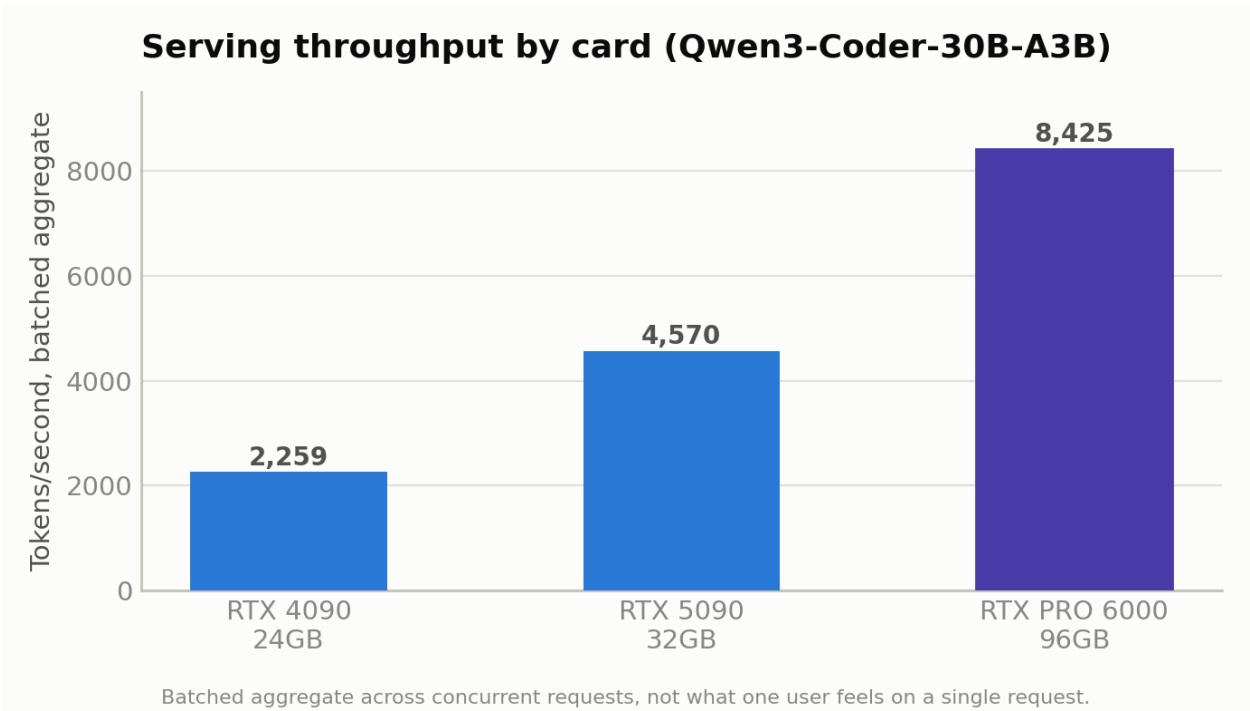
The two-5090 and one-PRO-6000 tiers are both defensible, not a strict upgrade path. Splitting *one* model across two consumer cards hits a PCIe bottleneck the PRO 6000's unified memory avoids — but that doesn't apply when the two cards run *different* models, which is this architecture (agent and image generation, separately). Two 5090s buys lane isolation; one PRO 6000 buys throughput per dollar and a single machine to operate. **Production note:** vLLM has no native Windows support — a Windows dev box is fine for building, but the machine that holds the real

GPU should run Ubuntu; plan that split before the card arrives, not after.

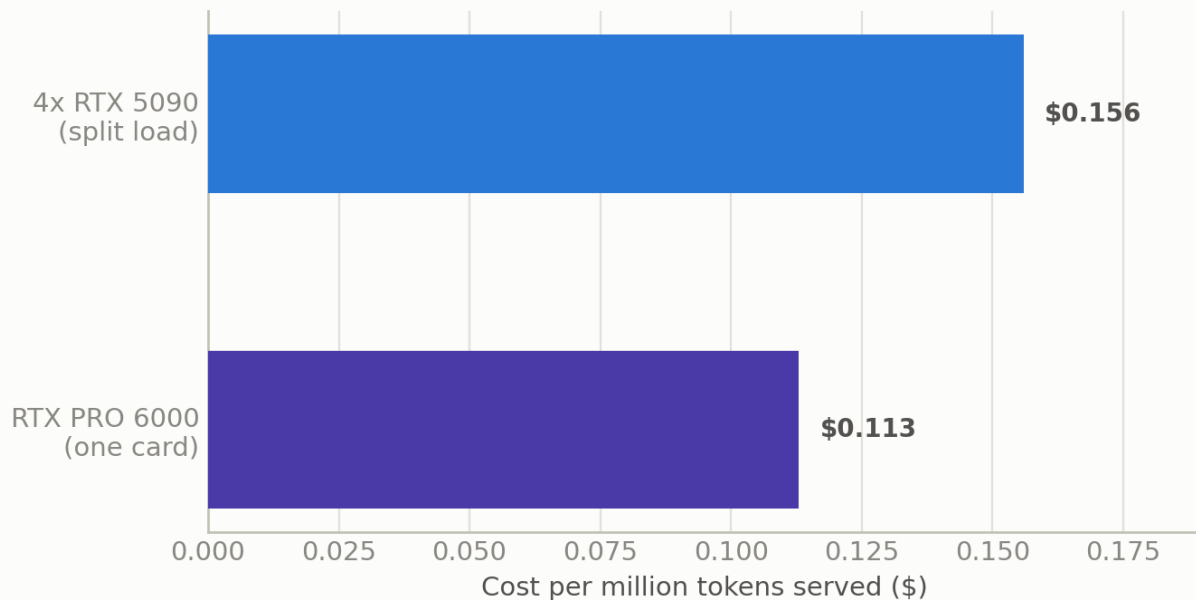
SERVING CAPACITY

What each card actually pushes

Measured serving Qwen3-Coder-30B-A3B with batching (CloudRift, September 2026). These are batched aggregate numbers across many concurrent requests — not the tokens/second one single user feels on their own request, which is far lower and dominated by the model's per-token latency rather than the card's total throughput.



Cost per token: one PRO 6000 beats four 5090s



Card	Throughput (batched)	Rental rate	Cost per token
RTX 4090 24 GB	2,259 tok/s	\$0.39/hr	—
RTX 5090 32 GB	4,570 tok/s	\$0.65/hr	—
RTX PRO 6000 96 GB	8,425 tok/s	\$1.29 – \$2.09/hr	\$0.113 / million tokens

One RTX PRO 6000 costs less per token served (\$0.113/million) than four RTX 5090s splitting the same load (\$0.156/million) — the throughput-per-dollar case for consolidating onto one large card once volume justifies it.

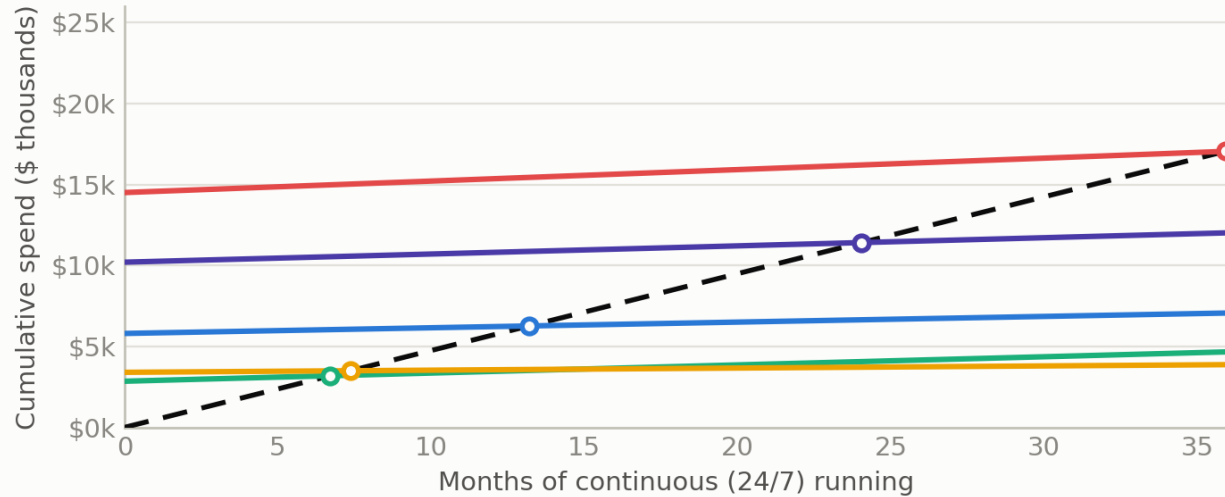
BUY VS. RENT

Where the lines cross

Every figure below assumes the alternative to buying is renting a 5090 continuously at \$0.65/hr — the cheapest capable rental tier. Owned-machine cost lines are not flat: they climb with that build's own electricity and cooling draw at DEWA commercial rates, which is why the "own" lines still slope gently upward.

— Rent a 5090, \$0.65/hr — Own: Strix Halo mini PC — Own: Two RTX 5090
 — Own: Two used RTX 3090 — Own: One RTX 5090 — Own: One RTX PRO 6000

Buying vs. renting: where the lines cross

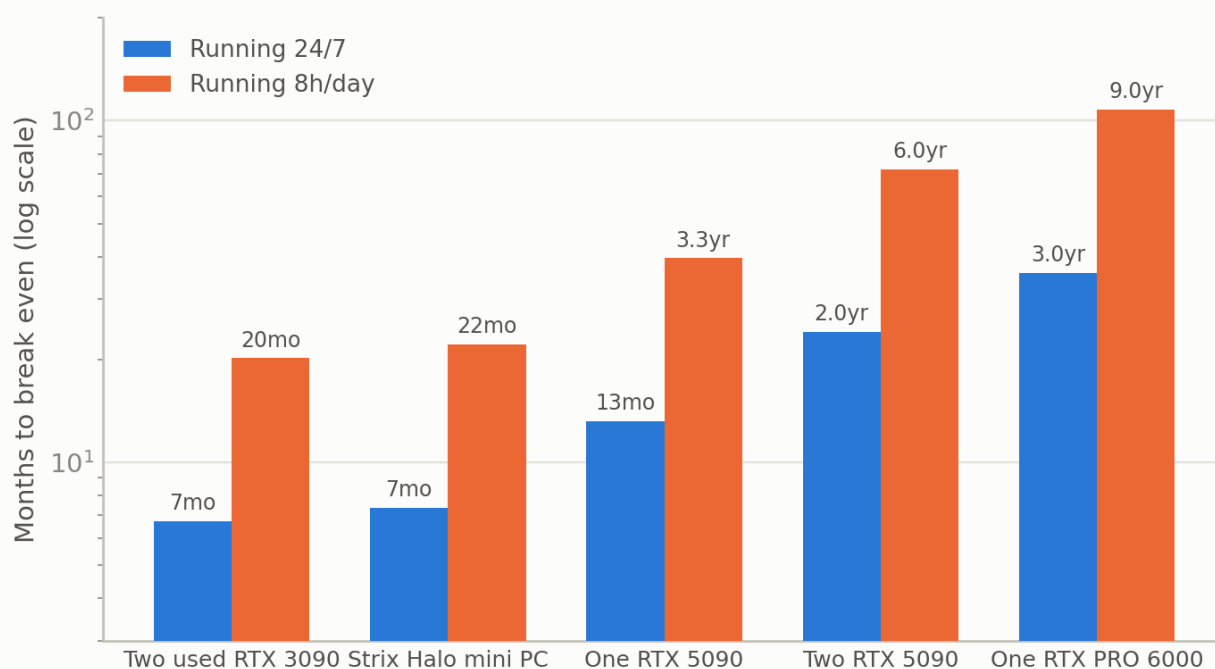


Circles mark break-even vs. renting a 5090. Own-cost lines include each build's electricity + cooling draw.

Build	Total cost	Break-even, 24/7	Break-even, 8h/day
Two used RTX 3090	\$2,850	6.7 months	20 months
Strix Halo mini PC	\$3,400	7.4 months	22 months
One RTX 5090	\$5,800	13.2 months	40 months
Two RTX 5090	\$10,200	24 months	6 years
One RTX PRO 6000	\$14,500	36 months	9 years

Continuous running is the most favourable case for buying. At eight hours a day — realistic for a prototype serving one person and one client — every crossing moves out by roughly three times, and nothing above the second row pays for itself inside three years.

How usage pattern moves the break-even point



Read this as a question about utilisation, not about hardware

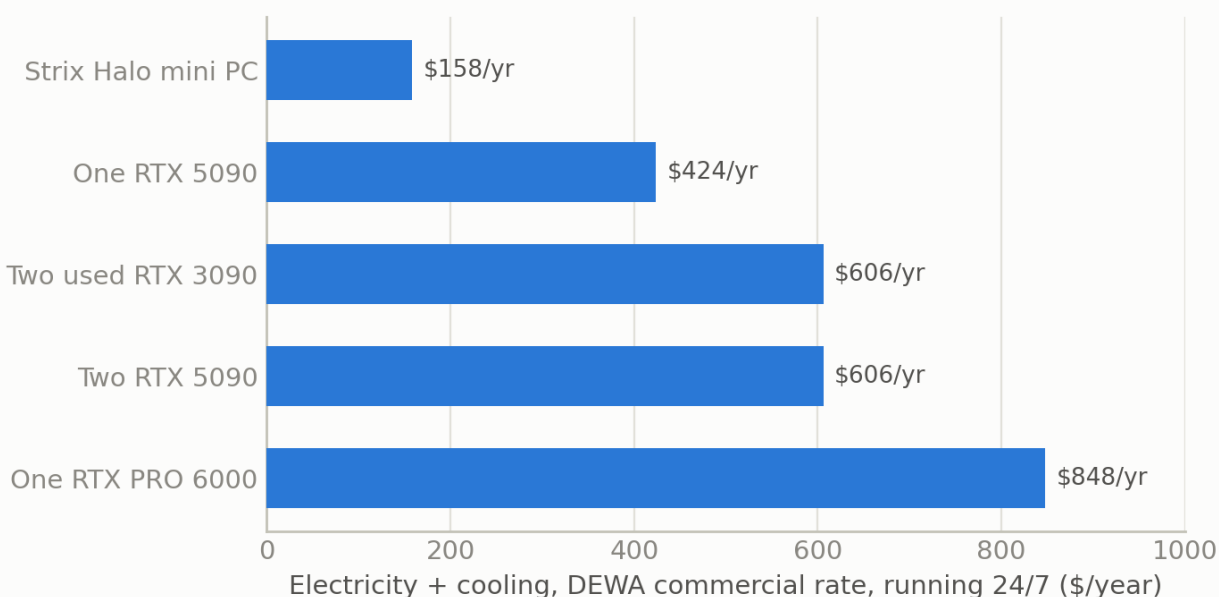
The cheapest mistake available here is renting for two months at about \$475 and learning what the workload actually needs. The most expensive is a \$14,500 machine sitting idle at 3am. **Buy when the machine will be busy; rent while still finding out.**

THE RUNNING COST

Electricity, and the cooling that follows it

DEWA commercial electricity runs 23–38 fils per kWh by tier plus a 6-fils fuel surcharge; figures here use a mid-tier blended rate of about 38 fils (\$0.104/kWh). Every watt a server draws leaves the box as heat, and a decent air-conditioning unit spends roughly one watt moving three back outside — so the "with cooling" column runs about a third higher than the electricity bill alone. Dubai's 230 V mains is a genuine advantage for multi-card builds: a 1,600 W supply for two 5090s sits comfortably on a normal circuit, where it would be near the limit of a household breaker on 120 V mains.

The running cost nobody quotes



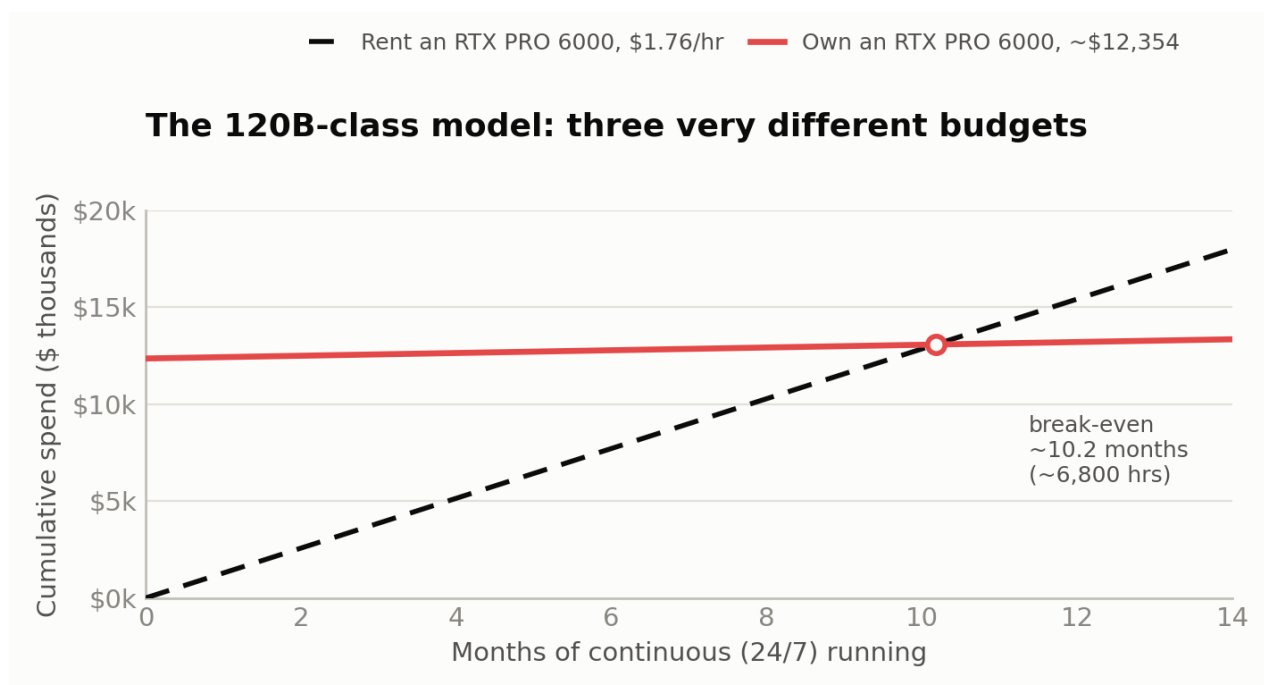
Build	Avg. draw	kWh/yr	Electricity	With cooling	Per hour owned
Strix Halo mini PC	130 W	1,139	\$118	\$158	\$0.018
One RTX 5090	350 W	3,066	\$319	\$424	\$0.048
Two used RTX 3090	500 W	4,380	\$456	\$606	\$0.069
Two RTX 5090	500 W	4,380	\$456	\$606	\$0.069
One RTX PRO 6000	700 W	6,132	\$638	\$848	\$0.097

It is not a large number in absolute terms — Dubai electricity is cheap by international standards. It matters because it is the line item that quietly turns a "free" always-on machine into roughly \$600–850 a year, and because it makes the low-power options genuinely cheaper to keep running, not only cheaper to buy.

A SPECIAL CASE

If the target is a 120B-class model

A dense 120B and a sparse (mixture-of-experts) 120B need completely different hardware, and the difference is the single biggest lever in this whole report. A dense 120B at Q4 is 71 GB of weights and needs a 96 GB-class card to hit roughly 20 tok/s — workstation or datacentre silicon, full stop. gpt-oss-120b, the likely candidate, is a mixture-of-experts model instead: 117B parameters total but only 5.1B active per token across 128 experts (4 active at a time), shipped natively at MXFP4. It needs the *memory* of a giant model and the *bandwidth* of a small one — reading only about 3 GB per token, which is roughly 24× faster than the dense model on the same hardware.



Off this scale entirely: a ~\$120 RAM upgrade runs gpt-oss-120b at 20-25 tok/s on the existing RTX 3060.

Path	Memory	Speed	Cost
RAM upgrade on the existing 3060	12 GB VRAM + 64 GB RAM	20 – 25 tok/s	~\$120
RTX PRO 6000 Blackwell (own)	96 GB	~20 tok/s (dense) / comfortable (sparse)	\$11,360 – \$13,349
Rent the same card	96 GB	same	\$1.76 – \$2.09/hr
Rent an H100 80 GB	80 GB	fastest option	\$2.89 – \$6.98/hr
Rent an A100 80 GB	80 GB	comfortable (sparse), tight at Q4 (dense)	from \$1.09/hr

The finding that changes the budget entirely

Someone has run gpt-oss-120b on an **RTX 4070 with 12 GB of VRAM** — the same class of card already installed — at over 25 tokens/second, by keeping attention on the GPU and pushing the expert layers into system RAM. The binding constraint wasn't the graphics card; it was having 64 GB of DDR5 to hold the experts in, clocked correctly. The current machine has 16 GB. **A roughly \$120 RAM upgrade is the cheapest path to a 120B model, not a new card** — provided the target is the sparse model. Buying a \$12,000 PRO 6000 only beats renting one after about 6,800 hours of continuous use, nine to ten months running day and night; that maths only closes once there is steady customer load, which is exactly the point at which the real requirement becomes clear.

Open question this depends on entirely: which 120B is actually meant. Dense vs. sparse changes the hardware answer from "a hundred-dollar RAM stick" to "a five-figure card," so naming the model is the highest-leverage next step before any of this budget is spent.

Three other shapes, and what each trades

Build	Price	Verdict
Strix Halo, 128 GB unified	\$2,600 – 5,000	Runs the recommended agent at ~72 tok/s on 140 W — but 256 GB/s bandwidth means dense models crawl (~5 tok/s), and there is no CUDA. Compelling as a low-power agent box; a risk behind an image-generating product.
Mac Studio M4 Max, 128 GB	\$3,999 – 5,999	546 GB/s, over twice the Strix Halo, ~180 W, excellent build quality and quiet. Same CUDA gap, more expensive, fixed configuration.
DGX Spark, 128 GB	\$4,699	Looks like the answer — NVIDIA, 128 GB, CUDA, one box — but its 273 GB/s bandwidth is lower than the RTX 3060 already owned. A capacity device, not a speed device; wrong axis for voice and image generation.

On buying used: a used RTX 3090 at \$700–850 gives 24 GB at 936 GB/s — over half a 5090's bandwidth for a fifth of the price. Two of them is the cheapest honest path to 48 GB, provided the seller can show power-on hours under roughly 20,000 and the card passes a stress test before payment; there is no warranty on used silicon.

The multimodal stack, if voice / vision / image / video are added

No single open model covers understanding, speech in, speech out, image generation and video generation at once. The chosen stack runs as five tools beside the existing document tools, coordinated by the tool-calling loop already built. Full stack resident is about 67 GB plus ~8 GB headroom for concurrent users — two 24 GB cards work if video is pinned to one; one 96 GB card covers everything.

Slot	Model	Licence	VRAM
Understanding (text/image/video/audio in)	MiniCPM-o 4.5, 9B	Apache 2.0	28 GB full duplex / 12 GB half
Speech in	Whisper Large V3 Turbo, 809M	MIT	6 GB
Speech out	Kokoro, 82M	Apache 2.0	2–3 GB, CPU-capable
Image generation	FLUX.2 [klein] 4B	Apache 2.0	~8 GB — the 9B sibling is non-commercial, do not swap it in
Video generation	LTX-2.3	Free under \$10M/yr revenue	24 GB at 720p fp8

Licence terms move fast: FLUX.2 klein-4B (Apache 2.0) and klein-9B (non-commercial) shipped the same day under the same family name — only the 4B is sellable. Read the LICENSE file in the exact repo before shipping anything in this table, rather than trusting the name alone.

A staged plan, cheapest first

1. Rent a 5090 for a month — about \$475.

Run the real stack against real documents and real clients if there are any yet. It costs less than the difference between two candidate builds and answers the only question that matters: how many people this actually has to serve.

2. Buy two used RTX 3090s and build around them — about \$2,850.

48 GB total, the fastest payback on this page at under seven months of continuous use, and enough headroom to run the agent on one card and image generation on the other. If one dies, it's an \$800 replacement rather than reconsidering a five-figure decision. It keeps the most options open.

3. Add 48–64 GB of system RAM before buying anything else for a 120B — about \$120.

Confirm the board takes DDR4/DDR5 at that capacity and that it clocks at its rated speed rather than the slower JEDEC default. This alone is the difference between not running a 120B and running one at conversational speed, if the target model is the sparse (MoE) one.

4. Move to an RTX PRO 6000 only when the machine is genuinely busy.

At eight hours a day it takes nine years to pay for itself. By the time load makes the maths close, the exact requirement — card count, memory, which model — will already be known from what came before.

Before spending anything

Name the actual 120B in mind — dense or MoE changes this from a hardware purchase into a memory upgrade. GPU prices in 2026 sit well above launch prices because of a memory shortage and are still moving; every price in this report should be re-checked before money changes hands, and the used market moves fastest of all.

REFERENCES

Where these numbers came from

- RTX PRO 6000 Blackwell pricing & RTX inference benchmarks — Thunder Compute / CloudRift, September 2026. Card prices, rental rates, 96 GB, 1.8 TB/s, batched throughput.
- Local AI workstation build guide 2026 — component costs by tier and the memory-shortage note.
- Custom AI workstation guide — multi-GPU platform requirements, PSU sizing, wall draw.
- Used GPU AI buying guide — second-hand prices, bandwidth, inspection checklist.
- AMD Strix Halo for local AI — price, measured bandwidth, tokens/second by model type.
- DGX Spark review; prebuilt AI workstations 2026.
- UAE business electricity tariffs — DEWA commercial slabs and fuel surcharge.
- Introducing gpt-oss (OpenAI) — total/active parameters, expert counts, MXFP4, the 80 GB figure.

- Optimizing gpt-oss-120b local inference — the RTX 4070 result, expert offloading, DDR5 clock discovery.
- Bandwidth and scaling figures measured directly on the project's own RTX 3060 (scripts/bench_models.py, 1 September 2026).

Full source links are kept in the two originating artifacts, "What The Machine Actually Costs" and "The Bandwidth Wall," and in the project roadmap memory.